Skillbit Technologies

Internship Project Report

Spam Email / SMS Classifier

Using NLP — TF-IDF + Naive Bayes

Domain: Machine Learning / NLP

Date: June 04, 2026

1. Objective

The objective of this project is to build a Natural Language Processing (NLP) based text classification system that can automatically detect and filter spam messages. The model is trained to classify SMS/email messages as either **Spam** or **Ham (Not Spam)** using TF-IDF feature extraction and probabilistic machine learning classifiers.

2. Dataset Details

Dataset Name	UCI SMS Spam Collection Dataset
Source	GitHub (raw CSV) with built-in fallback
Total Samples	5,572 messages
Spam Messages	747 (~13.4%)
Ham Messages	4,825 (~86.6%)
Class Balance	Imbalanced — handled via stratified split
Format	Tab-separated: label, message text

3. Text Preprocessing

Raw text data was cleaned through the following preprocessing pipeline before being passed to the feature extractor:

Step	Operation	Example
1	Lowercase conversion	WIN CASH! → win cash!
2	Remove digits/numbers	Call 09061 now → Call now
3	Remove punctuation	win cash!! → win cash
4	Strip extra whitespace	win cash → win cash

4. Feature Extraction — TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency) was used to convert raw text into numerical feature vectors. It assigns higher weights to words that are important in a specific document but rare across the corpus — making it ideal for spam detection.

max_features	5,000 (top frequent terms)

ngram_range	(1, 2) — unigrams and bigrams
stop_words	English stopwords removed
Output Shape	4,457 train x 5,000 features

5. Algorithms Used

Two classifiers were trained and compared:

Model	Description	Accuracy
Naive Bayes (MultinomialNB)	Probabilistic classifier based on Bayes' theorem. Assumes feature independence. Ideal for text	98.21%
Logistic Regression	Linear classifier that models the log-odds of each class. Uses sigmoid function for binary classi	96.59%

6. Results & Performance

98.21% Best Accuracy (Naive Bayes)	**0.98** Ham Precision	**0.98** Spam Precision
1.00 Ham Recall	**0.89** Spam Recall	**1115** Test Samples

Naive Bayes — Per-Class Performance:

Class	Precision	Recall	F1-Score	Support
Ham (Not Spam)	0.98	1.00	0.99	966
Spam	0.98	0.89	0.93	149
Weighted Avg	0.98	0.98	0.98	1115

7. Key Findings

- Naive Bayes outperforms Logistic Regression — a common result in NLP text classification tasks.
- Top spam keywords identified: free, txt, stop, claim, prize, won, mobile, reply.
- TF-IDF with bigrams captures two-word patterns like 'free entry', 'win cash', 'call now'.

- Class imbalance (13% spam) was handled effectively using stratified train-test split.
- The model achieves near-perfect Ham recall (1.00) — very few legitimate messages are blocked.
- Live prediction demo correctly classified all 4 test messages with high confidence.

8. Technology Stack

Language	Python 3.x
ML Library	scikit-learn (MultinomialNB, LogisticRegression, TfidfVectorizer)
Text Processing	re, string (built-in Python modules)
Data	pandas, numpy
Visualization	matplotlib, seaborn
IDE	Visual Studio Code

9. Conclusion

The Spam Email/SMS Classifier project successfully demonstrates the application of Natural Language Processing techniques for text classification. Using TF-IDF vectorization and the Multinomial Naive Bayes algorithm, the model achieved an impressive accuracy of 98.21% on 1,115 test samples. The project covered the complete NLP pipeline — text preprocessing, feature extraction, model training, comparative evaluation, and live prediction. The high accuracy and near-perfect Ham recall make this model suitable for real-world spam filtering applications.